

Program : Diploma in Manufacturing Technology	
Course Code : 2701	Course Title: Basic Production Process
Semester : 2	Credits: 3
Course Category: Engineering Science	
Periods per week: 3 (L:2 T:1 P:0)	Periods per semester: 45

Course Objectives:

- To provide good knowledge about the importance of studying basic production process in daily life and industry
- To familiarize with the basics of engineering materials and different tools used in mechanical workshop.
- To familiarize with different basic production processes such as fitting, metal joining, foundry etc.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Science		Applied Physics I	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize ferrous and non ferrous metals and their alloys	8	Understanding
CO2	Identify different tools in bench work and fitting, and carpentry	9	Understanding
CO3	Describe different patterns, pattern making , and foundry tools and operations	12	Understanding
CO4	Explain various smithy and forging operations, welding process and sheet metal operations.	14	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7
CO1	2	-	-	-	-	-	-
CO2	2	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-
CO4	2	-	-	-	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize ferrous and non ferrous metals and their alloys		
M1.01	List different Ferrous metals and explain making of pig iron and cast iron	2	Understanding
M1.02	Explain Bessemer process and identify different types of steels based on carbon content	2	Understanding
M1.03	List different non ferrous metals and general properties	2	Understanding
M1.04	Identify alloys of copper, zinc, tin, lead and aluminium	2	Understanding
Contents: Ferrous metals -introduction-pig iron- blast furnace- cast iron-types of cast iron - cupola furnace- wrought iron- steel -Bessemer process-types of steel based on carbon content-low carbon steel-medium carbon steel-high carbon steel. Non ferrous metals – introduction – properties – aluminium, zinc, copper, tin, lead-alloys – duralumin – brass – bronze - gun metal – Alnico - muntz metal			
CO2	Identify different tools in bench work and fitting, and carpentry		
M2.01	Identify different vices used in fitting	2	Understanding
M2.02	List types of chisels and chipping process	2	Understanding
M2.03	Identify different types of files and filing process and different marking tools.	3	Understanding

M2.04	Understand the basics in carpentry.	2	Understanding
	Series Test – I	1	
Contents: Fitting and Bench work: Vices and their types-filing-methods of filing-classification of files-hammers-chisels and their classification-marking tools-scribers-punches-v block-angle plate-measuring tools-hacksaws-blades-setting of teeth-scrapers-drills and reamers-taps and dies. Carpentry- Different tools used in carpentry-carpentry joints.			
CO3	Describe different patterns, pattern making, and foundry tools and operations		
M3.01	Identify different pattern materials and explain types of patterns and pattern making allowances	3	Understanding
M3.02	Explain different foundry tools and equipments	3	Understanding
M3.03	Understand different moulding sands and its properties	3	Understanding
M3.04	Identify different types of core and casting processes	3	Understanding
Contents: Pattern making and foundry- Pattern-pattern materials-wood-metal-plastics-rubbers-plasters-waxes-types of patterns-solid pattern-split-match plate-cope and drag pattern-gated-sweep pattern-Pattern making allowances-shrinkage-draft-machining-distortion-rapping. Foundry- Tools and Equipments-moulding boxes-moulding sands-types of moulding sand-properties-sand preparation-types of moulding process-bench and floor moulding-pit moulding-sweep moulding-plate moulding-machine moulding-investment moulding-cores-core boxes-types-chaplets-casting-types (name only)-risers-function-chillers.			
CO4	Explain various smithy and forging operations, welding process and sheet metal operations.		
M 4.01	Understanding smithing and forging tools	3	Understanding
M 4.02	Identify different operations in smithing and forging	3	Understanding
M 4.03	Explain welding and different types of welding	5	Understanding
M 4.04	Identify different sheet metal operations	3	Understanding
	Series Test – II	1	

Contents:

Smithy and forging-Introduction-forgability-forging temperature-heating devices-tools-anvil-swage block-hand rammers-hammers-tongs-chisels-swages-fullers-flatters-punch and drift-smith forging operations-upsetting-bending-drawing down-welding-setting down-cutting-punching-fullering.

Welding-introduction-weldability-types of welding-arc welding-polarity-dc and ac arc welding-electrodes-fluxes-arc welding methods-MIG-TIG-Metal arc-Gas welding-oxy acetylene gas welding-types of flames-equipments-resistance welding-types-submerged arc welding-atomic hydrogen welding-thermit welding-Soldering-Brazing-defects in welding

Sheet metal operations-metals-tools used-operations-shearing-drawing-bending-squeezing.

Text / Reference

T/R	Book Title/Author
T1	Elements of Work Shop Technology- Vol. I – S. K. Hajra Choudhury
T2	Elements of Work Shop Technology- Vol. II – S. K. Hajra Choudhury
T3	A text book of production Technology – P C Sharma
R4	A text book of production engineering – P C Sharma
R5	Production Technology – HMT
R6	Machine Tools and Tool Design- P.C Sharma(S Chand)

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/112/104/112104195/
2	https://bharatskills.gov.in/Home/StudyMaterial?var=vNuH8QjfGrhOzGTIvfa6cA==&Default=YES